

# MAKERERE UNIVERSITY

## Weekly Progress Report - Bee Hive Audio Monitoring Project

|                      |                                  |                       |                     |
|----------------------|----------------------------------|-----------------------|---------------------|
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| <b>Project Area:</b> | Bee Hive Audio Monitoring        | <b>Report Period:</b> | This Week           |
| <b>Focus:</b>        | Audio cleaning and tool practice | <b>Date:</b>          | 12 June 2026        |

### 1. Introduction

This weekly report presents the progress made on the Bee Hive Audio Monitoring Project. The main goal of this stage was to build a clear understanding of the project, learn how bee audio data can be prepared, and gain practical experience with audio-cleaning tools before moving deeper into coding and analysis.

### 2. Work Completed This Week

- Read project materials and carried out background research to understand the purpose of the bee hive monitoring project, especially the use of sound signals in detecting bee behaviours.
- Learned how to clean audio files by identifying and removing unwanted noise such as birds, strong wind, knocking sounds on the hive, and other background disturbances.
- Gained practical knowledge on how to use audio tools, especially Audacity and Adobe Audition, for waveform and spectrum viewing, noise selection, amplitude reduction, and audio editing.
- Successfully cleaned queen quacking audio samples and started organising cleaned sections into smaller, categorised audio groups for future analysis.

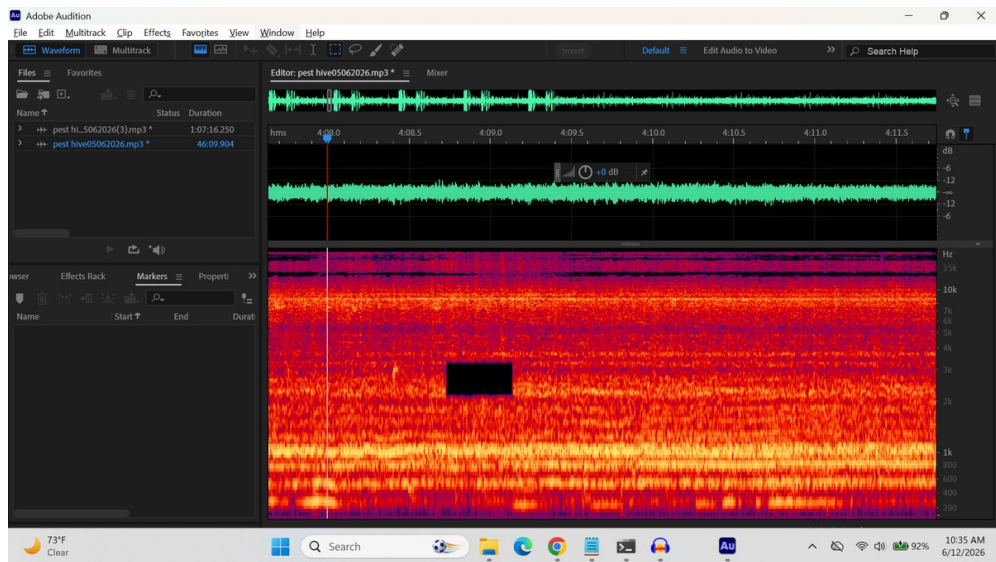
### 3. Skills and Knowledge Gained

- Understanding audio features: I improved my ability to observe audio signals using waveform and spectrogram views.
- Noise identification: I learned how to recognise unwanted parts in audio recordings, including bird sounds, sharp knocks, and wind interference.
- Audio cleaning methods: I practised deleting unwanted sections, reducing amplitude using the brush tool, and removing rectangular noise patterns visible in the spectrogram.
- Tool confidence: I became more comfortable using Audacity and Adobe Audition, which will help me work faster in the coming weeks.

### 4. Methods Practised

The attached workflow document shows the practical methods used during audio cleaning. Page 1 shows understanding of audio features and removing unwanted noise. Page 2 shows deleting unwanted sections, reducing amplitude using the brush, and deleting rectangular bird-like noise. Page 3 shows cutting out small cleaned and categorised audio groups for future processing.

Sample workflow evidence:



## 5. Challenges and Observations

- At the beginning, it took time to understand the tools and how to adjust audio views, especially when switching between time, waveform, and frequency/spectrum views.
- Some noises were difficult to remove because they overlapped with important bee sounds. This made it necessary to clean carefully so that useful information was not destroyed.
- Working with audio requires patience because every recording may contain different types of background noise.

## 6. Plan for Next Week

- Clean more bee audio files at a faster pace because I am now getting used to Audacity and Adobe Audition.
- Continue practising noise reduction, amplitude reduction, and selection of useful audio segments.
- Carry out more research on how cleaned audio files can be used in code for feature extraction and machine learning.
- Begin connecting the cleaned audio workflow to the coding stage by studying how to load audio files, extract features, and prepare them for analysis.

## 7. Conclusion

This week was important for building the foundation of the Bee Hive Audio Monitoring Project. I gained a clearer understanding of the project, learned practical audio-cleaning techniques, and became more confident using Audacity and Adobe Audition. In the next week, I expect to improve my speed in cleaning more audio files and strengthen my understanding of how the cleaned audio can be used in code for project development.

## Appendix: Additional Workflow Evidence

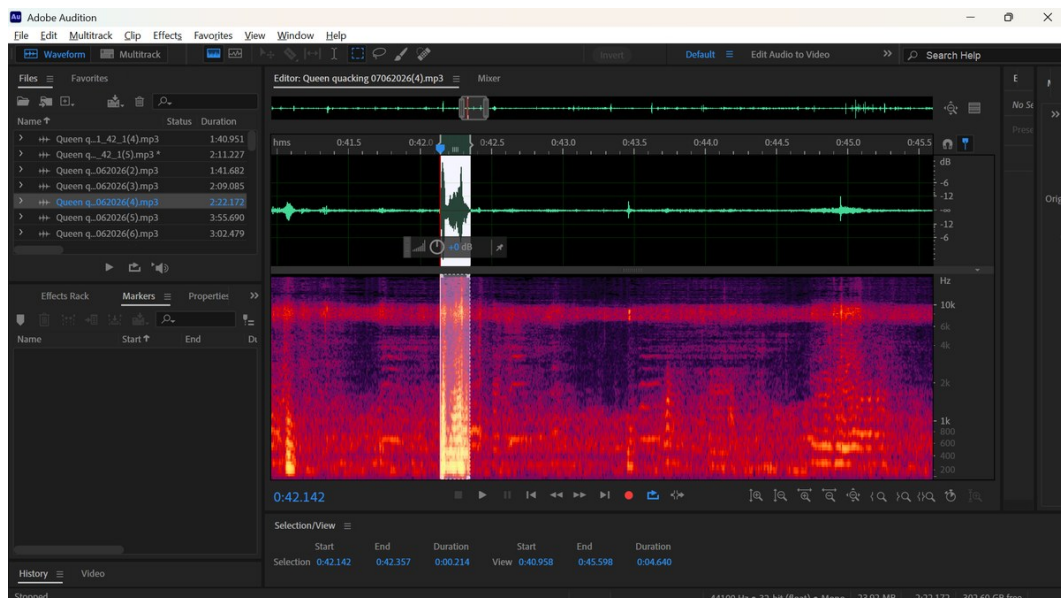


Figure 2: Audio cleaning workflow screenshot

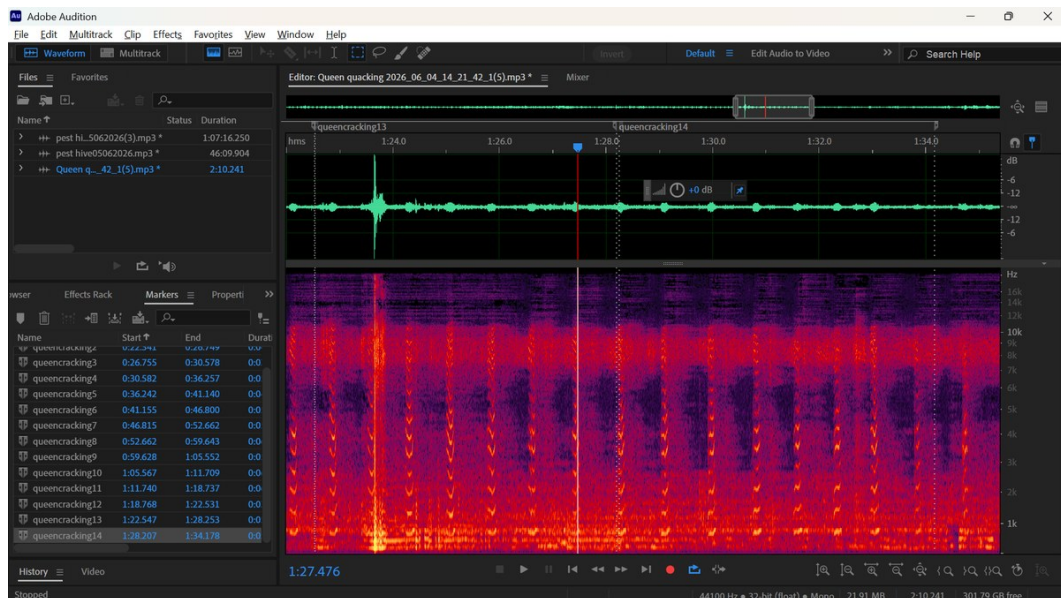


Figure 3: Audio cleaning workflow screenshot

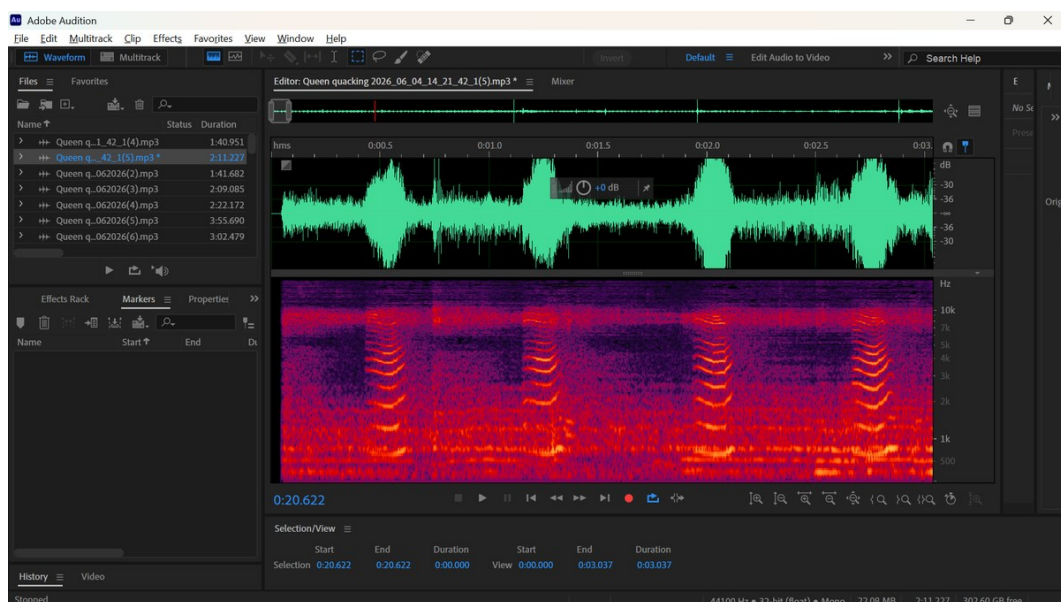


Figure 4: Audio cleaning workflow screenshot

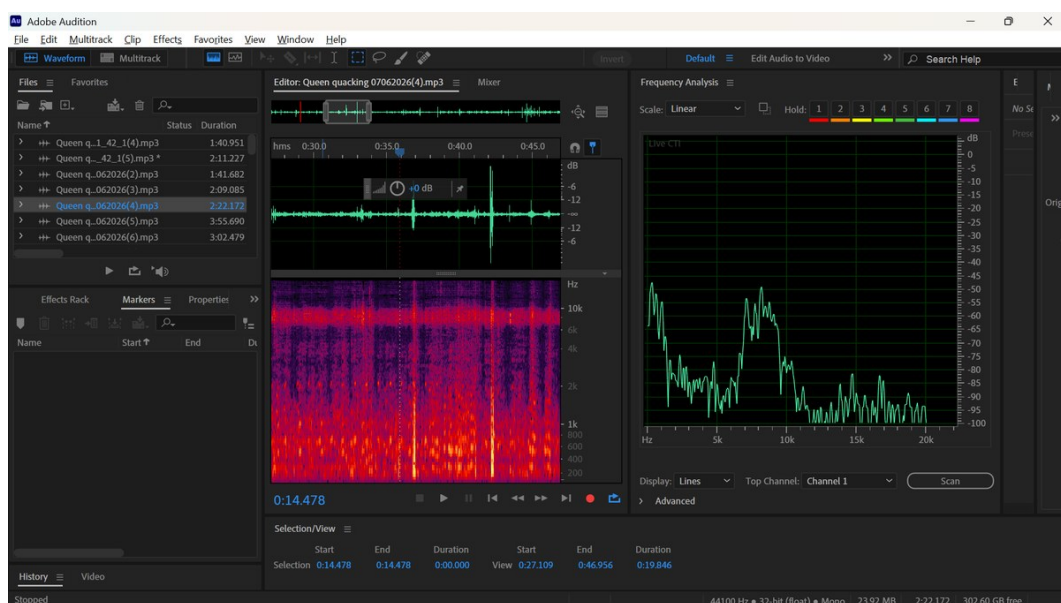


Figure 5: Audio cleaning workflow screenshot

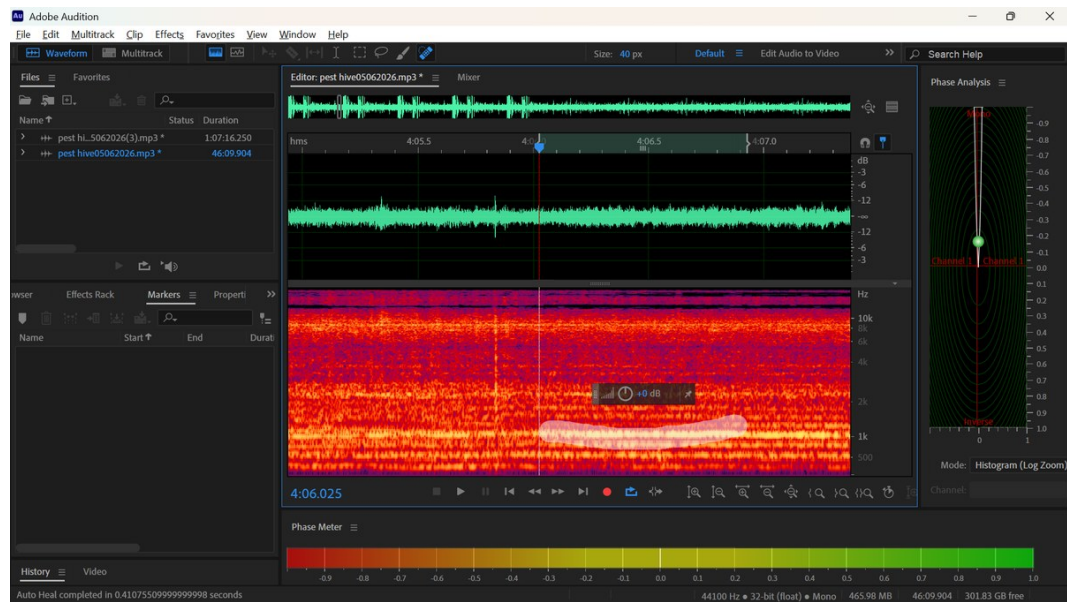


Figure 6: Audio cleaning workflow screenshot